



R3.2 Electron transfer reactions

Reactivity 3 · What are the mechanisms of chemical change? · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Oxidation is loss of electrons (increase in oxidation state) and reduction is gain of electrons (decrease in oxidation state); redox reactions involve simultaneous oxidation and reduction, and can be balanced using half-equations, with the reducing agent being oxidised (and reducing something else) and the oxidising agent being reduced. Electrochemical (voltaic/galvanic) cells convert chemical energy from a spontaneous redox reaction into electrical energy, using two half-cells connected by a wire (external circuit, for electron flow) and a salt bridge/porous barrier (to complete the circuit and maintain electrical neutrality), with electrons flowing from the anode (oxidation, negative electrode) to the cathode (reduction, positive electrode) in the external circuit. Electrolytic cells use electrical energy to drive a non-spontaneous redox reaction (electrolysis), for example to decompose a molten ionic compound or aqueous solution into its elements.

Worked example: follow the reasoning

Balance the redox reaction between manganate(VII) ions and iron(II) ions in acidic solution, given half-equations: $\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$, and $\text{Fe}^{2+} \rightarrow \text{Fe}^{3+} + \text{e}^-$.

Step 1: Electrons lost must equal electrons gained; the Mn half-equation involves 5 electrons, the Fe half-equation involves 1 electron.

Step 2: Multiply the Fe half-equation by 5 so both involve 5 electrons: $5\text{Fe}^{2+} \rightarrow 5\text{Fe}^{3+} + 5\text{e}^-$.

Step 3: Add the two half-equations and cancel the 5e^- on each side: $\text{MnO}_4^- + 8\text{H}^+ + 5\text{Fe}^{2+} \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O} + 5\text{Fe}^{3+}$.

Answer: $\text{MnO}_4^-(\text{aq}) + 8\text{H}^+(\text{aq}) + 5\text{Fe}^{2+}(\text{aq}) \rightarrow \text{Mn}^{2+}(\text{aq}) + 4\text{H}_2\text{O}(\text{l}) + 5\text{Fe}^{3+}(\text{aq})$.

Quick check · multiple choice

In the reaction $\text{Zn}(\text{s}) + \text{Cu}^{2+}(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{Cu}(\text{s})$, which statement is correct?

- A Zn is reduced and acts as the oxidising agent
- B Cu^{2+} is reduced and acts as the oxidising agent
- C Zn^{2+} is oxidised and acts as the reducing agent
- D Cu is oxidised and acts as the reducing agent

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A voltaic cell is constructed using a $\text{Zn}(\text{s})/\text{Zn}^{2+}(\text{aq})$ half-cell and a $\text{Cu}(\text{s})/\text{Cu}^{2+}(\text{aq})$ half-cell connected by a wire and a salt bridge. [7]
 - (a) Identify which electrode is the anode and which is the cathode, and state the half-equation occurring at each. [4]
 - (b) State the direction of electron flow in the external circuit. [1]
 - (c) State the purpose of the salt bridge in this cell. [2]

Final answer: _____



Answers and explanation

Question 1 [7]

(a) Anode (oxidation, negative electrode): Zn, half-equation $\text{Zn(s)} \rightarrow \text{Zn}^{2+}(\text{aq}) + 2\text{e}^-$. Cathode (reduction, positive electrode): Cu, half-equation $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu(s)}$. [4: correct identity and correct half-equation for each electrode]

(b) Electrons flow through the external circuit from the zinc electrode (anode) to the copper electrode (cathode). [1]

(c) The salt bridge allows ions to migrate between the two half-cells to complete the circuit and maintain electrical neutrality in each half-cell (as ions are produced/consumed), without allowing the bulk solutions to mix directly. [2]

Quick check answer

B. Cu^{2+} gains electrons (reduced, oxidation state $+2 \rightarrow 0$) and causes Zn to be oxidised, so Cu^{2+} is the oxidising agent.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Chemistry HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.